

Muhammad Hammad Sarwar

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Summary

Analytical and backend-focused Computer Science undergraduate at COMSATS University Islamabad with a strong foundation in Python, distributed backend architectures, and RESTful API design. Proven track record of building scalable microservices, high-throughput asynchronous systems using Redis, RabbitMQ, and Celery, and integrating AI vision models into full-stack workflows. Certified by Harvard, IBM, and the University of Michigan, combining rigorous algorithmic problem-solving with hands-on production engineering, database modeling, and clean system design.

Experience

Maxenius Private Limited

Python Developer Intern | July 2026 – Present

- Engineered and optimized 15+ REST API endpoints in Python, reducing median server response latency by 35% through query optimization and database indexing.
- Structured modular service layers and relational schemas using Alembic migration workflows and Pydantic validation, eliminating invalid payload errors across 100% of incoming requests.
- Enforced stateless JWT authentication and granular Role-Based Access Control (RBAC), securing 20+ protected endpoints against unauthorized privilege escalation.
- Standardized API documentation via OpenAPI/Swagger and resolved 25+ critical backend defects across production release cycles, improving sprint delivery velocity by 20%.

Projects

Distributed E-Commerce & Flash Sale Engine

Backend Architect | July 2026 – August 2026

- Architected a high-concurrency flash sale engine using Python (Flask), PostgreSQL, Redis, RabbitMQ, and Celery to manage high-traffic checkout spikes.
- Configured Redis in-memory caching and distributed locking to prevent race conditions, inventory overselling, and database bottlenecks.
- Offloaded asynchronous order processing, invoice generation, and notifications to background workers using RabbitMQ and Celery.
- Optimized PostgreSQL transactional operations and database indexing, cutting read/write latency under peak load by 40%.

LogicFlow AI

Lead Developer & Architect | February 2026

- Spearheaded a full-stack engineering application using TypeScript and React Flow for interactive, node-based circuit schematic visualization.
- Integrated Gemini Multimodal Vision Models via a Python backend to parse visual data from hand-drawn circuit sketches into digital layouts.
- Developed custom algorithmic evaluation engines to translate parsed structural layouts into production-ready, functional Verilog code and digital simulations.
- Optimized component rendering and modular state management pipelines, reducing interface re-rendering lag by 25%.
- Programmed real-time gate logic evaluation algorithms to validate truth tables and propagation delays across complex digital circuits.

HR & Team Management System

Python & Backend Developer | July 2026

- Built a comprehensive HR and team management REST API using Python, Flask, and PostgreSQL for role-based workforce administration.
- Deployed secure JSON Web Token (JWT) authentication, password hashing, and granular Role-Based Access Control (RBAC).
- Modeled relational schemas for department hierarchies, employee profiles, attendance, and leave management with strict data validation.
- Authored automated unit and integration tests across 30+ endpoints to validate API contracts, role permissions, and edge cases.

Education

COMSATS University Islamabad

Lahore, Pakistan

Bachelor of Science, Computer Science

September 2024 – July 2028

Relevant Coursework: Data Structures, Design & Analysis of Algorithms, Database Systems, Object-Oriented Programming, Computer Networks, Software Engineering, Artificial Intelligence, Discrete Structures

Board Of Intermediate and Secondary Education Faisalabad

Faisalabad, Pakistan

FSc, Pre-Medical

August 2020 – September 2023

Graduated with Grade A

Relevant Coursework: Physics, Chemistry, Biology, Mathematics

Certifications & Courses

- **Harvard University:** CS50x Puzzle Day 2025
- **IBM:** Git & GitHub, HTML/CSS/JavaScript, Software Engineering (Honors), Cloud Computing
- **University of Michigan:** Python for Everybody Specialization (Coursera)
- **DeepLearning.AI:** AI For Everyone
- **Microsoft & HP LIFE:** PowerPoint Productivity, Effective Presentations

Technical Skills

- **Languages:** Python, TypeScript, JavaScript, SQL, C++, Java, HTML5, CSS3
- **Frameworks & Backend:** Flask, FastAPI, Celery, Node.js, Express, React, Next.js, Tailwind CSS
- **Databases & Caching:** PostgreSQL, MySQL, Redis, MongoDB, SQLite
- **Distributed Systems & DevOps:** RabbitMQ, Docker, Git, GitHub, REST APIs, Postman, Linux
- **Architecture & Concepts:** System Design, Microservices, Asynchronous Task Processing, Distributed Locking, JWT/RBAC, Data Structures & Algorithms, OOP